

Name : MD.Sheebaaz 2303A51823 Batch : 26

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| <b>SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE</b>                            |                                                                                    | <b>DEPARTMENT OF COMPUTER SCIENCE ENGINEERING</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| <b>Program Name:</b> B. Tech                                                             |                                                                                    | <b>Assignment Type:</b> Lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Academic Year:</b> 2025-2026 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| <b>Course Coordinator Name</b>                                                           |                                                                                    | Dr. Rishabh Mittal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| <b>Instructor(s)Name</b>                                                                 |                                                                                    | <table border="1"> <tr><td>Mr. S Naresh Kumar</td><td></td></tr> <tr><td>Ms. B. Swathi</td><td></td></tr> <tr><td>Dr. Sasanko Shekhar Gantayat</td><td></td></tr> <tr><td>Mr. Md Sallauddin</td><td></td></tr> <tr><td>Dr. Mathivanan</td><td></td></tr> <tr><td>Mr. Y Srikanth</td><td></td></tr> <tr><td>Ms. N Shilpa</td><td></td></tr> <tr><td>Dr. Rishabh Mittal (Coordinator)</td><td></td></tr> <tr><td>Dr. R. Prashant Kumar</td><td></td></tr> <tr><td>Mr. Ankushavali MD</td><td></td></tr> <tr><td>Mr. B Viswanath</td><td></td></tr> <tr><td>Ms. Sujitha Reddy</td><td></td></tr> <tr><td>Ms. A. Anitha</td><td></td></tr> <tr><td>Ms. M.Madhuri</td><td></td></tr> <tr><td>Ms. Katherashala Swetha</td><td></td></tr> <tr><td>Ms. Velpula sumalatha</td><td></td></tr> <tr><td>Mr. Bingi Raju</td><td></td></tr> <tr><td>Mr. G. Kranthi</td><td></td></tr> </table> |                                 | Mr. S Naresh Kumar |  | Ms. B. Swathi |  | Dr. Sasanko Shekhar Gantayat |  | Mr. Md Sallauddin |  | Dr. Mathivanan |  | Mr. Y Srikanth |  | Ms. N Shilpa |  | Dr. Rishabh Mittal (Coordinator) |  | Dr. R. Prashant Kumar |  | Mr. Ankushavali MD |  | Mr. B Viswanath |  | Ms. Sujitha Reddy |  | Ms. A. Anitha |  | Ms. M.Madhuri |  | Ms. Katherashala Swetha |  | Ms. Velpula sumalatha |  | Mr. Bingi Raju |  | Mr. G. Kranthi |  |
| Mr. S Naresh Kumar                                                                       |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Ms. B. Swathi                                                                            |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Dr. Sasanko Shekhar Gantayat                                                             |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Mr. Md Sallauddin                                                                        |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Dr. Mathivanan                                                                           |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Mr. Y Srikanth                                                                           |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Ms. N Shilpa                                                                             |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Dr. Rishabh Mittal (Coordinator)                                                         |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Dr. R. Prashant Kumar                                                                    |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Mr. Ankushavali MD                                                                       |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Mr. B Viswanath                                                                          |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Ms. Sujitha Reddy                                                                        |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Ms. A. Anitha                                                                            |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Ms. M.Madhuri                                                                            |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Ms. Katherashala Swetha                                                                  |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Ms. Velpula sumalatha                                                                    |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Mr. Bingi Raju                                                                           |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| Mr. G. Kranthi                                                                           |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| <b>Course Code</b>                                                                       | 23CS002PC304                                                                       | <b>Course Title</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AI Assisted Coding              |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| <b>Year/Sem</b>                                                                          | III/I                                                                              | <b>Regulation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | R23                             |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| <b>Date and Day of Assignment</b>                                                        | Week 5 - Thursday                                                                  | <b>Time(s)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 23CSBTB01 To 23CSBTB52          |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| <b>Duration</b>                                                                          | 2 Hours                                                                            | <b>Applicable to Batches</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | All Batches                     |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| <b>AssignmentNumber:</b> 9.4 (Present assignment number)/24(Total number of assignments) |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
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| <b>Q.No.</b>                                                                             | <b>Question</b>                                                                    | <b>ExpectedTime to complete</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |
| 1                                                                                        | <b>Lab 9 – Documentation Generation: Automatic Documentation and Code Comments</b> | Week 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                 |                    |  |               |  |                              |  |                   |  |                |  |                |  |              |  |                                  |  |                       |  |                    |  |                 |  |                   |  |               |  |               |  |                         |  |                       |  |                |  |                |  |

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|  | <p><b>Lab Objectives</b></p> <ul style="list-style-type: none"> <li>• To use AI-assisted coding tools for generating Python documentation and code comments.</li> <li>• To apply zero-shot, few-shot, and context-based prompt engineering for documentation creation.</li> <li>• To practice generating and refining docstrings, inline comments, and module-level documentation.</li> <li>• To compare outputs from different prompting styles for quality analysis.</li> </ul> <p><b>Lab Outcomes</b></p> <ul style="list-style-type: none"> <li>• Generate structured code documentation using AI tools</li> <li>• Apply appropriate documentation styles to different code contexts</li> <li>• Improve code readability through selective commenting</li> <li>• Convert informal developer comments into professional documentation</li> <li>• Analyze and refine AI-generated documentation.</li> </ul>                                                                                                                                                                                                                                                                                                                                          |  |
|  | <p><b>Task 1: Auto-Generating Function Documentation in a Shared Codebase</b></p> <p><b>Scenario</b><br/> You have joined a development team where several utility functions are already implemented, but the code lacks proper documentation. New team members are struggling to understand how these functions should be used.</p> <p><b>Task Description</b><br/> You are given a Python script containing multiple functions without any docstrings.</p> <p>Using an AI-assisted coding tool:</p> <ul style="list-style-type: none"> <li>• Ask the AI to automatically generate <b>Google-style function docstrings</b> for each function</li> <li>• Each docstring should include: <ul style="list-style-type: none"> <li>◦ A brief description of the function</li> <li>◦ Parameters with data types</li> <li>◦ Return values</li> <li>◦ At least one example usage (if applicable)</li> </ul> </li> </ul> <p>Experiment with different prompting styles (zero-shot or context-based) to observe quality differences.</p> <p><b>Expected Outcome</b></p> <ul style="list-style-type: none"> <li>• A Python script with well-structured Google-style docstrings</li> <li>• Docstrings that clearly explain function behavior and usage</li> </ul> |  |

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|  | <ul style="list-style-type: none"> <li>• Improved readability and usability of the codebase</li> </ul> <pre> code: def add_numbers(a, b):     """Adds two numbers together.     .     .     .     Args:     .   a (int or float): The first number.     .   b (int or float): The second number.     .     .     Returns:     .   int or float: The sum of a and b.     .     .     Example:     .   &gt;&gt;&gt; add_numbers(3, 5)     .       8     .   &gt;&gt;&gt; add_numbers(1.5, 2.3)     .       3.8     .   """     .   return a + b     .     .     def multiply_numbers(x, y):     .   """Multiplies two numbers.     .     .   This function takes two numerical inputs and returns their     .   product.     .     .   Args:     .       x (int or float): The first number to multiply.     .       y (int or float): The second number to multiply.     .     .   Returns:     .       int or float: The product of x and y.     .     .   Example:     .       &gt;&gt;&gt; multiply_numbers(4, 2)     .           8     .       &gt;&gt;&gt; multiply_numbers(2.5, 3)     .           7.5     .       """     .       return x * y     .     .     def greet_user(name, greeting='Hello'):     .   """Greet a user with a customizable greeting.     .     .     .   Args: </pre> |  |
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|  | <ul style="list-style-type: none"> <li>• name (str): The name of the user to greet.</li> <li>• greeting (str, optional): The greeting message. Defaults to 'Hello'.</li> <li>•</li> <li>• Returns:</li> <li>• str: A complete greeting message.</li> <li>•</li> <li>• Example:</li> <li>• &gt;&gt;&gt; greet_user('Alice')</li> <li>• 'Hello, Alice!'</li> <li>• &gt;&gt;&gt; greet_user('Bob', greeting='Hi')</li> <li>• 'Hi, Bob!'</li> <li>• """</li> <li>• return f'{greeting}, {name}!'</li> <li>•</li> <li>• Above are example functions with complete Google-style docstrings, including descriptions, parameter types, return values, and example usage where applicable.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
|  | <p><b>Task 2: Enhancing Readability Through AI-Generated Inline Comments</b></p> <p><b>Scenario</b><br/>A Python program contains complex logic that works correctly but is difficult to understand at first glance. Future maintainers may find it hard to debug or extend this code.</p> <p><b>Task Description</b><br/>You are provided with a Python script containing:</p> <ul style="list-style-type: none"> <li>• Loops</li> <li>• Conditional logic</li> <li>• Algorithms (such as Fibonacci sequence, sorting, or searching)</li> </ul> <p>Use AI assistance to:</p> <ul style="list-style-type: none"> <li>• Automatically insert <b>inline comments only for complex or non-obvious logic</b></li> <li>• Avoid commenting on trivial or self-explanatory syntax</li> </ul> <p>The goal is to improve clarity without cluttering the code.</p> <p><b>Expected Outcome</b></p> <ul style="list-style-type: none"> <li>• A Python script with concise, meaningful inline comments</li> <li>• Comments that explain <i>why</i> the logic exists, not <i>what</i> Python syntax does</li> <li>• Noticeable improvement in code readability</li> </ul> <p>code:<br/>def add_numbers(a, b):</p> |  |

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|  | <ul style="list-style-type: none"> <li>• """Adds two numbers together.</li> <li>•</li> <li>• Args:</li> <li>• a (int or float): The first number.</li> <li>• b (int or float): The second number.</li> <li>•</li> <li>• Returns:</li> <li>• int or float: The sum of a and b.</li> <li>•</li> <li>• Example:</li> <li>• &gt;&gt;&gt; add_numbers(3, 5)</li> <li>• 8</li> <li>• &gt;&gt;&gt; add_numbers(1.5, 2.3)</li> <li>• 3.8</li> <li>• """</li> <li>• return a + b</li> <li>•</li> <li>• def multiply_numbers(x, y):</li> <li>• """Multiplies two numbers.</li> <li>•</li> <li>• This function takes two numerical inputs and returns their product.</li> <li>•</li> <li>• Args:</li> <li>• x (int or float): The first number to multiply.</li> <li>• y (int or float): The second number to multiply.</li> <li>•</li> <li>• Returns:</li> <li>• int or float: The product of x and y.</li> <li>•</li> <li>• Example:</li> <li>• &gt;&gt;&gt; multiply_numbers(4, 2)</li> <li>• 8</li> <li>• &gt;&gt;&gt; multiply_numbers(2.5, 3)</li> <li>• 7.5</li> <li>• """</li> <li>• return x * y</li> <li>•</li> <li>• def greet_user(name, greeting='Hello'):</li> <li>• """Greets a user with a customizable greeting.</li> <li>•</li> <li>• Args:</li> <li>• name (str): The name of the user to greet.</li> <li>• greeting (str, optional): The greeting message. Defaults to 'Hello'.</li> <li>•</li> </ul> |  |
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|  | <ul style="list-style-type: none"> <li>• Returns:</li> <li>• str: A complete greeting message.</li> <li>•</li> <li>• Example:</li> <li>• <code>&gt;&gt;&gt; greet_user('Alice')</code></li> <li>• <code>'Hello, Alice!'</code></li> <li>• <code>&gt;&gt;&gt; greet_user('Bob', greeting='Hi')</code></li> <li>• <code>'Hi, Bob!'</code></li> <li>• <code>"""</code></li> <li>• <code>return f'{greeting}, {name}!'</code></li> <li>•</li> <li>• The above code block contains the functions. As explained, no inline comments were added because the logic is straightforward and already well-documented by the Google-style docstrings, adhering to your request to only comment on 'complex or non-obvious logic' and 'without commenting on trivial syntax'.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |
|  | <p><b>Task 3: Generating Module-Level Documentation for a Python Package</b></p> <p><b>Scenario</b><br/>Your team is preparing a Python module to be shared internally (or uploaded to a repository). Anyone opening the file should immediately understand its purpose and structure.</p> <p><b>Task Description</b><br/>Provide a complete Python module to an AI tool and instruct it to automatically generate a <b>module-level docstring</b> at the top of the file that includes:</p> <ul style="list-style-type: none"> <li>• The purpose of the module</li> <li>• Required libraries or dependencies</li> <li>• A brief description of key functions and classes</li> <li>• A short example of how the module can be used</li> </ul> <p>Focus on clarity and professional tone.</p> <p><b>Expected Outcome</b></p> <ul style="list-style-type: none"> <li>• A well-written multi-line module-level docstring</li> <li>• Clear overview of what the module does and how to use it</li> <li>• Documentation suitable for real-world projects or repositories</li> </ul> <p>code:</p> <pre>"""This module provides a collection of utility functions for basic arithmetic operations and user greetings.</pre> <ul style="list-style-type: none"> <li>•</li> <li>• Purpose:</li> </ul> |  |

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|  | <ul style="list-style-type: none"> <li>• This module aims to demonstrate the implementation of clear, well-documented Python functions</li> <li>• using Google-style docstrings, module-level documentation, and selective inline comments.</li> <li>• It serves as an example of best practices for code documentation.</li> <li>•</li> <li>• Required Libraries:</li> <li>• This module does not explicitly require any external libraries beyond Python's standard built-in functions.</li> <li>•</li> <li>• Main Components: <ul style="list-style-type: none"> <li>• - `add_numbers(a, b)`: A function to add two numerical inputs.</li> <li>• - `multiply_numbers(x, y)`: A function to multiply two numerical inputs.</li> <li>• - `greet_user(name, greeting)`: A function to generate a personalized greeting message.</li> </ul> </li> <li>•</li> <li>• Minimal Example: <ul style="list-style-type: none"> <li>• To use the functions in this module, simply import them and call them as needed:</li> <li>•</li> <li>• <pre>&gt;&gt;&gt; from my_module import add_numbers, multiply_numbers, greet_user</pre></li> <li>• <pre>&gt;&gt;&gt; sum_result = add_numbers(10, 5)</pre></li> <li>• <pre>&gt;&gt;&gt; print(f"Sum: {sum_result}")</pre></li> <li>• <pre>Sum: 15</pre></li> <li>•</li> <li>• <pre>&gt;&gt;&gt; product_result = multiply_numbers(4, 6)</pre></li> <li>• <pre>&gt;&gt;&gt; print(f"Product: {product_result}")</pre></li> <li>• <pre>Product: 24</pre></li> <li>•</li> <li>• <pre>&gt;&gt;&gt; greeting_message = greet_user("Alice", "Hello")</pre></li> <li>• <pre>&gt;&gt;&gt; print(greeting_message)</pre></li> <li>• <pre>Hello, Alice!</pre></li> <li>•</li> <li>• <pre>&gt;&gt;&gt; default_greeting = greet_user("Bob")</pre></li> <li>• <pre>&gt;&gt;&gt; print(default_greeting)</pre></li> <li>• <pre>Hello, Bob!</pre></li> <li>• <pre>"""</pre></li> <li>• The above code block contains the functions. As explained, no inline comments were added because the logic is straightforward and already well-documented by the Google-style docstrings, adhering to your request to only comment on 'complex or non-obvious logic' and 'without commenting on trivial syntax'.</li> </ul> </li></ul> |  |
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| <div data-bbox="354 193 1299 231" data-label="Section-Header"><h2>Task 4: Converting Developer Comments into Structured Docstrings</h2></div> <div data-bbox="354 285 496 323" data-label="Section-Header"><h3>Scenario</h3></div> <div data-bbox="354 325 1282 434" data-label="Text"><p>In a legacy project, developers have written long explanatory comments inside functions instead of proper docstrings. The team now wants to standardize documentation.</p></div> <div data-bbox="354 476 625 518" data-label="Section-Header"><h3>Task Description</h3></div> <div data-bbox="354 516 1247 592" data-label="Text"><p>You are given a Python script where functions contain detailed inline comments explaining their logic.</p></div> <div data-bbox="354 590 496 623" data-label="Text"><p>Use AI to:</p></div> <div data-bbox="401 625 1265 770" data-label="List-Group"><ul style="list-style-type: none"><li>• Automatically convert these comments into structured <b>Google-style or NumPy-style docstrings</b></li><li>• Preserve the original meaning and intent of the comments</li><li>• Remove redundant inline comments after conversion</li></ul></div> <div data-bbox="354 814 654 854" data-label="Section-Header"><h3>Expected Outcome</h3></div> <div data-bbox="401 854 1299 1839" data-label="List-Group"><ul style="list-style-type: none"><li>• Functions with clean, standardized docstrings</li><li>• Reduced clutter inside function bodies</li><li>• Improved consistency across the codebase</li></ul><p>code:</p><pre>"""This module provides a collection of utility functions for basic arithmetic operations and user greetings.</pre><ul style="list-style-type: none"><li>•</li><li>• Purpose:</li><li>• This module aims to demonstrate the implementation of clear, well-documented Python functions</li><li>• using Google-style docstrings, module-level documentation, and selective inline comments.</li><li>• It serves as an example of best practices for code documentation.</li><li>•</li><li>• Required Libraries:</li><li>• This module does not explicitly require any external libraries beyond Python's standard built-in functions.</li><li>•</li><li>• Main Components:</li><li>• - `add_numbers(a, b)`: A function to add two numerical inputs.</li><li>• - `multiply_numbers(x, y)`: A function to multiply two numerical inputs.</li><li>• - `greet_user(name, greeting)`: A function to generate a personalized greeting message.</li><li>•</li><li>• Minimal Example:</li></ul></div> |  |
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|  | <p><b>Task 5: Building a Mini Automatic Documentation Generator</b></p> <p><b>Scenario</b><br/>Your team wants a simple internal tool that helps developers start documenting new Python files quickly, without writing documentation from scratch.</p> <p><b>Task Description</b><br/>Design a small Python utility that:</p> <ul style="list-style-type: none"> <li>• Reads a given <code>.py</code> file</li> <li>• Automatically detects: <ul style="list-style-type: none"> <li>◦ Functions</li> <li>◦ Classes</li> </ul> </li> <li>• Inserts <b>placeholder Google-style docstrings</b> for each detected function or class</li> </ul> <p>AI tools may be used to assist in generating or refining this utility.<br/>Note: The goal is <b>documentation scaffolding</b>, not perfect documentation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |

## Expected Outcome

- A working Python script that processes another .py file
- Automatically inserted placeholder docstrings
- Clear demonstration of how AI can assist in documentation automation
- Code:

```
import ast
import re

def _get_indentation(line):
    """Returns the leading whitespace of a line."""
    match = re.match(r'^(\s*)', line)
    return match.group(1) if match else ""

def _generate_function_docstring(func_node, indent):
    """Generates a placeholder Google-style docstring for a function."""
    docstring_lines = []

    docstring_lines.append(indent + " " + "'''" + "Summary of the function's purpose.")

    # Args
    all_args = list(func_node.args.args) + list(func_node.args.kwonlyargs)
    if all_args or func_node.args.vararg or func_node.args.kwarg:
        docstring_lines.append(f'{indent}"')
        docstring_lines.append(f'{indent}  Args:")
        for arg in all_args:
            docstring_lines.append(f'{indent}    {arg.arg} (type): Description of {arg.arg}.")
        if func_node.args.vararg:
            docstring_lines.append(f'{indent}    *{func_node.args.vararg.arg} (type): Description of variable arguments.")
        if func_node.args.kwarg:
            docstring_lines.append(f'{indent}    **{func_node.args.kwarg.arg} (type): Description of keyword arguments.")

    # Returns
    docstring_lines.append(f'{indent}"')
    docstring_lines.append(f'{indent}  Returns:")
    docstring_lines.append(f'{indent}    type: Description of the return value.")

    docstring_lines.append(indent + " " + "''")
    return "\n".join(docstring_lines) + "\n"
```

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|  | <ul style="list-style-type: none"> <li>•</li> <li>• <code>def _generate_class_docstring(class_node, indent):</code></li> <li>• <code>    """Generates a placeholder Google-style docstring for a</code></li> <li>• <code>    class."""</code></li> <li>• <code>    docstring_lines = []</code></li> <li>•</li> <li>• <code>    docstring_lines.append(indent + " " + """ + "Summary of</code></li> <li>• <code>    the class's purpose.")</code></li> <li>• <code>    docstring_lines.append(f'{indent}')</code></li> <li>• <code>    docstring_lines.append(f'{indent} Attributes:')</code></li> <li>• <code>    docstring_lines.append(f'{indent} # Add class attributes</code></li> <li>• <code>    here, e.g.:')</code></li> <li>• <code>    docstring_lines.append(f'{indent} # attribute_name (type):</code></li> <li>• <code>    Description of the attribute.")</code></li> <li>• <code>    docstring_lines.append(indent + " " + """)</code></li> <li>• <code>    return "\n".join(docstring_lines) + "\n"</code></li> <li>•</li> <li>• <code>def generate_placeholder_docstrings(file_path):</code></li> <li>• <code>    """</code></li> <li>• <code>    Reads a Python file, detects functions and classes without</code></li> <li>• <code>    docstrings,</code></li> <li>• <code>    and returns the file content with placeholder Google-style</code></li> <li>• <code>    docstrings inserted.</code></li> <li>•</li> <li>• <code>    Args:</code></li> <li>• <code>        file_path (str): The path to the input Python file.</code></li> <li>•</li> <li>• <code>    Returns:</code></li> <li>• <code>        str: The modified file content with placeholder docstrings.</code></li> <li>• <code>    """</code></li> <li>• <code>    with open(file_path, 'r', encoding='utf-8') as f:</code></li> <li>• <code>        lines = f.readlines()</code></li> <li>•</li> <li>• <code>    tree = ast.parse("".join(lines))</code></li> <li>•</li> <li>• <code>    # Collect insertions: (line_number_to_insert_after,</code></li> <li>• <code>    docstring_text)</code></li> <li>• <code>    insertions = []</code></li> <li>•</li> <li>• <code>    for node in ast.walk(tree):</code></li> <li>• <code>        if isinstance(node, (ast.FunctionDef, ast.ClassDef)):</code></li> <li>• <code>            # Check if a docstring already exists</code></li> <li>• <code>            if ast.get_docstring(node) is None:</code></li> <li>• <code>                # Get the indentation of the definition line</code></li> <li>• <code>                line_idx = node.lineno - 1 # ast.lineno is 1-based, so</code></li> <li>• <code>                adjust for 0-based list index</code></li> </ul> |  |
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|  | <pre> •         original_indent = _get_indentation(lines[line_idx]) • •         docstring_to_insert = "" •         if isinstance(node, ast.FunctionDef): •             docstring_to_insert = _generate_function_docstring(node, original_indent) •         elif isinstance(node, ast.ClassDef): •             docstring_to_insert = _generate_class_docstring(node, original_indent) • •         # Insert after the definition line (node.lineno is 1-based) •         insertions.append((node.lineno, docstring_to_insert)) • •         # Sort insertions by line number in reverse to avoid shifting issues during insertion •         insertions.sort(key=lambda x: x[0], reverse=True) • •         modified_lines = list(lines) •         for line_num_1based, docstring_text in insertions: •             # list.insert() uses 0-based index, so line_num_1based is already the correct index for insertion after the definition line. •             modified_lines.insert(line_num_1based, docstring_text) • •         return "".join(modified_lines) • • # --- Example Usage (Uncomment to run) --- • # # Create a dummy Python file for demonstration • # dummy_code_content = """ • # import os • # • # class MyClass: • #     pass • # • # def func_without_docstring(a, b, *args, kwarg1=None, **kwargs): • #     pass • # • # def another_func_with_docstring(x): • #     """This function already has a docstring.""" • #     return x * 2 • # • # class AnotherClassWithDocstring: • #     """This class already has a docstring.""" • #     def __init__(self): • #         pass • # </pre> |  |
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|  | <ul style="list-style-type: none"> <li>• <code>## A simple function to be ignored</code></li> <li>• <code># def simple_function():</code></li> <li>• <code>#     return 1</code></li> <li>• <code># """</code></li> <li>• <code>#</code></li> <li>• <code># dummy_file_path = 'example_code.py'</code></li> <li>• <code># with open(dummy_file_path, 'w') as f:</code></li> <li>• <code>#     f.write(dummy_code_content)</code></li> <li>• <code>#</code></li> <li>• <code># print(f'Original content of {dummy_file_path}:\n---\n{dummy_code_content}---\n')</code></li> <li>• <code>#</code></li> <li>• <code>## Generate and print the modified content</code></li> <li>• <code># modified_output =</code><br/><code>generate_placeholder_docstrings(dummy_file_path)</code></li> <li>• <code># print(f'Modified content with placeholder docstrings:\n---\n{modified_output}')</code></li> <li>• <code>#</code></li> <li>• <code>## Clean up the dummy file</code></li> <li>• <code># import os</code></li> <li>• <code># os.remove(dummy_file_path)</code></li> <li>•</li> <li>•</li> </ul> <p>The code block above contains the <code>generate_placeholder_docstrings</code> function and its helpers. You can uncomment the 'Example Usage' section at the bottom of the code cell to see it in action. It will:</p> <ul style="list-style-type: none"> <li>• Create a temporary <code>example_code.py</code> file.</li> <li>• Call <code>generate_placeholder_docstrings</code> on this file.</li> <li>• Print the original and the modified content with the new docstrings inserted.</li> <li>• Clean up the temporary file.</li> </ul> <p><b>Note: Report should be submitted a word document for all tasks in a single document with prompts, comments &amp; code explanation, and output and if required, screenshots</b></p> |  |
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